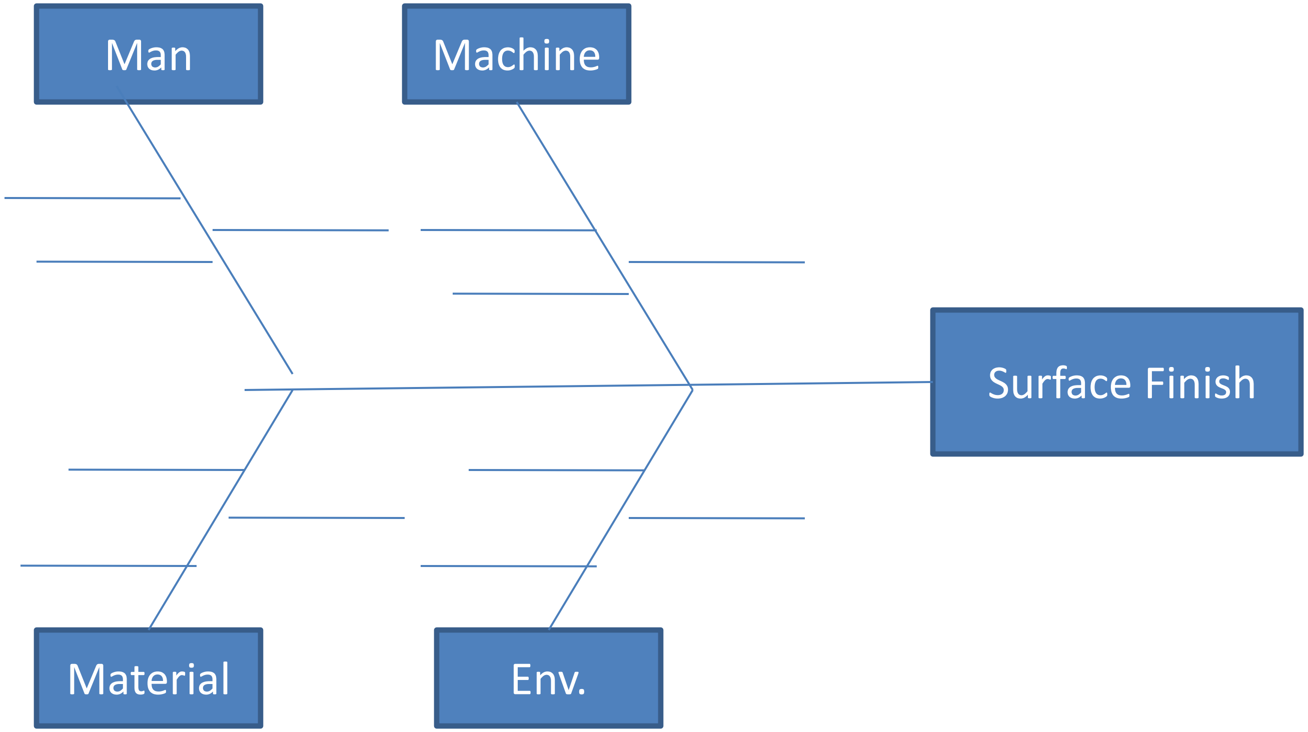
Design of Experiments (DOE)

A statistical methodology to systematically investigate a systems input – output relationship.

In other words, it is used to find cause-and-effect relationships. This information is needed to manage process inputs in order to optimize the output.

Design of Experiments (DOE)

1. Cause and Effect diagram



Robust Design

Design of Experiments (DOE)

2. Select important factors for study (Process parameters)

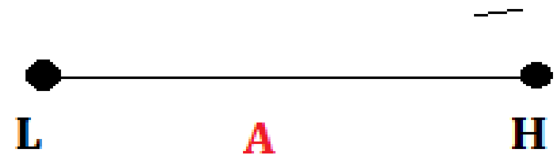
a. Speed (H L)
b. Feed (H L)
c. Type of tool (Carbide(C) HSS (S))

Factors

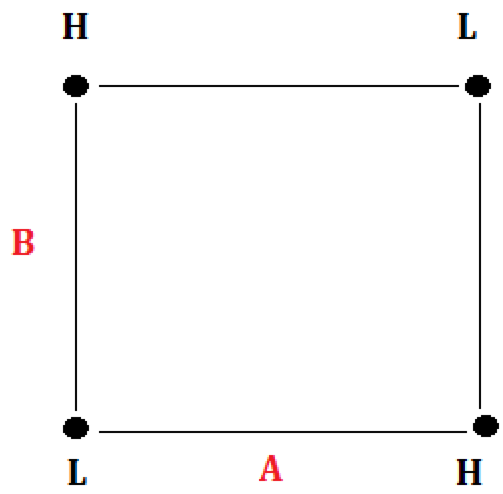
Levels

Design of Experiments (DOE)

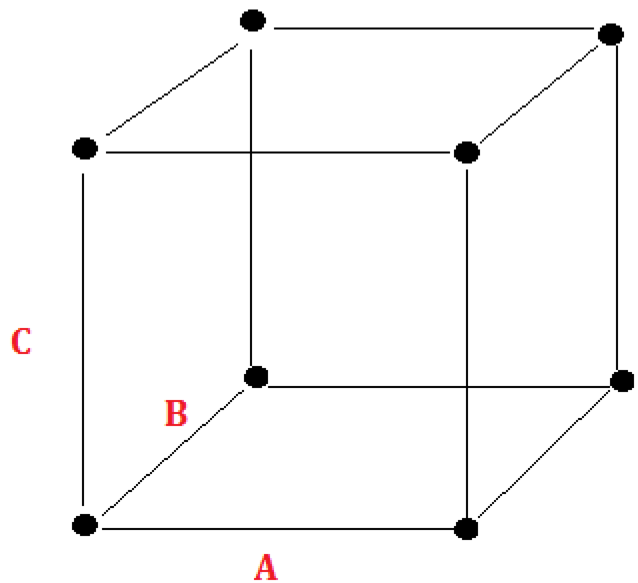
2 Level Designs $2^1 = 2$ runs



$2^2 = 4$ runs



$2^3 = 8$ Run



2^2 Factorial Design

Ex. No	Speed	Feed	S X F
1	1	1	2
2	2	1	1
3	1	2	1
4	2	2	2

Design of Experiments (DOE)

3. Plan for experiments:

Ex.No	Speed	Feed	Tool	1	2	3
1	H	H	C			
2	L	H	C			
3	H	L	C			
4	L	L	C			
5	H	H	S			
6	L	H	S			
7	H	L	S			
8	L	L	S			

Design of Experiments (DOE)

4. Full factorial design:
(Include interactions)

Ex. No	Speed	Feed	Tool	S X F	F X T	S X T	S X T X F
1	1	1	1	2			1
2	2	1	1	1			2
3	1	2	1	1			2
4	2	2	1	2			1
5	1	1	2	2			2
6	2	1	2	1			1
7	1	2	2	1			1
8	2	2	2	2			2

Balanced

Factors	MF	2 IF	3 IF	4 IF
2	2	1		
3	3	3	1	
4	4	6	4	1

Contrast and Effect of factors

Treatment	Speed	Feed	1	2	3
1	250	0.1	10	11	9
2	300	0.1	13	12	12
3	250	0.2	14	15	17
4	300	0.2	8	4	5

Speed (A) → 250 (-), 300 (+)

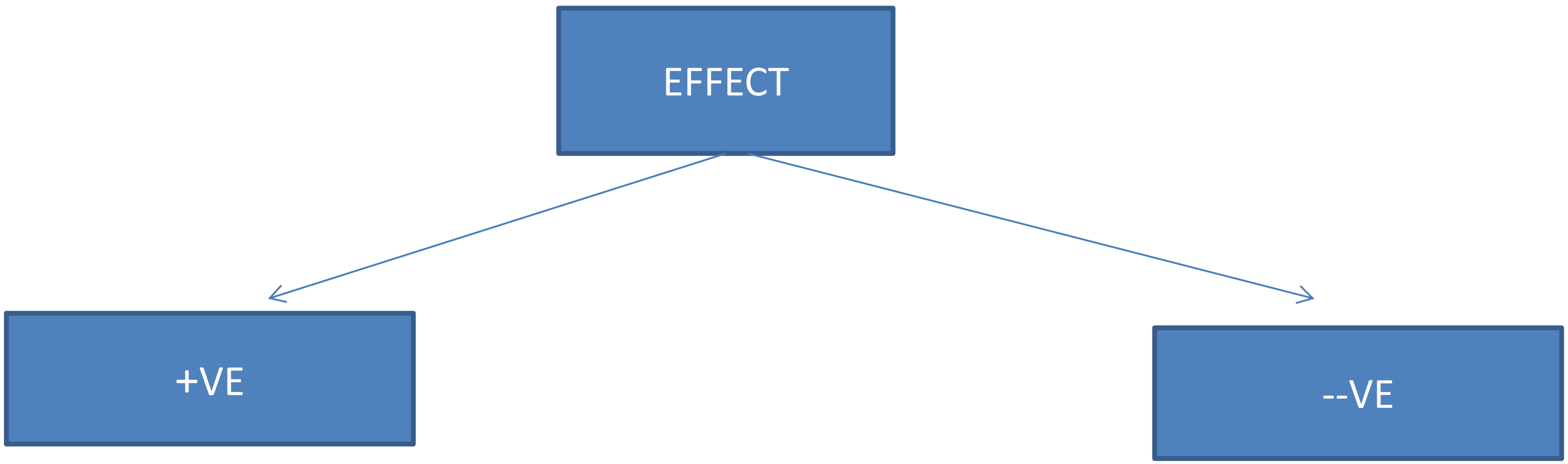
Feed (B) → 0.1(-), 0.2 (+)

r = 3 (replicates)

Treatment	A	B	AB	1	2	3	Total
(1)	-	-	+	10	11	9	30
a	+	-	-	13	12	12	37
b	-	+	-	14	15	17	46
ab	+	+	+	8	4	5	17

Contrast = (Sum of responses at High) – (Sum of responses at Low)

Effect= (Contrast) / ($r * 2^{k-1}$) for 2^k factorial design



LEVEL	**RESPONSE**
HIGH	HIGH
LOW	LOW

LEVEL	**RESPONSE**
HIGH	LOW
LOW	HIGH

Treatment	1	2	3	Total
(1)	10	11	9	30
a	13	12	12	37
b	14	15	17	46
ab	8	4	5	17

Main Effect of Factor A = **[** a + ab – ((1) + b)**]** / r x 2^{k-1} = (37+17-30-46) /6 = -3.66

Main Effect of Factor B = **[** b + ab – ((1) + a)**]** / r x 2^{k-1} = (46+17-30-37) /6 = -0.66

Effect of Interaction AB = **[** (1) + ab – (a + b)**]** / r x 2^{k-1} = (30+17-37-46) /6 = -6

A+ & B+ or A- & B -

A +

B+

ANOVA

H_0 for A = Response is not influenced by A

H_0 for B = Response is not influenced by B

H_0 = Response is not influenced by AXB interaction

Treatment	1	2	3	Total
(1)	10	11	9	30
a	13	12	12	37
b	14	15	17	46
ab	8	4	5	17
				130

Sum of Squares: SS

$$= \frac{(\text{Contrast})^2}{r \times 2^k}$$

Contrast = (Sum of responses at High) – (Sum of responses at Low)
Effect= (Contrast) / (r * 2^{k-1}) for 2^k factorial design (k : no. of factors)

Sum of Squares: $SS_{A/B/AB}$

$$= \frac{(\text{Contrast})^2}{r \times 2^k}$$

$$SS_T = \left\{ \sum \sum X_{ij}^2 \right\} - (T^2/N)$$

$$SS_E = SS_T - (SS_A + SS_B + SS_{AB})$$

Mean Square: MS

$$= \frac{SS}{DOF}$$

$$F_0 = MS/MS_E$$

Sum of Squares:

- $SS_A = (37+17-30-46)^2/(3 \text{ x } 2^2) = 40.33$
- $SS_B = (46+17-30-37)^2/(3 \text{ x } 2^2) = \quad 1.33$
- $SS_{AB} = (30+17-37-46)^2/(3 \text{ x } 2^2) = 108$
- $SS_T = [10^2+11^2+9^2+13^2+\ldots\ldots.+5^2] - (130^2/N)$

  $= 1574 - 1408.33 = 165.67$

- $SS_E = SS_T - (SS_A + SS_B + SS_{AB}) = 165.67 - (40.33+1.33+108)$

  $= 16.01$

Source of Variation	SS	DOF	MS= SS/DOF	F_0= MS/MS_E
A	40.33	1	40.33	20.165
B	1.33	1	1.33	0.665
AB	108	1	108	54
Error	16.01	8	2	
Total	165.67	11	15.1	

For **p value** or **α** value of **0.05**

Df1 of 1 (Numerator)

Df2 of 8 (Denominator)

We get $F^*_{\alpha} = F^*_{0.05,1,8} = \mathbf{5.32}$

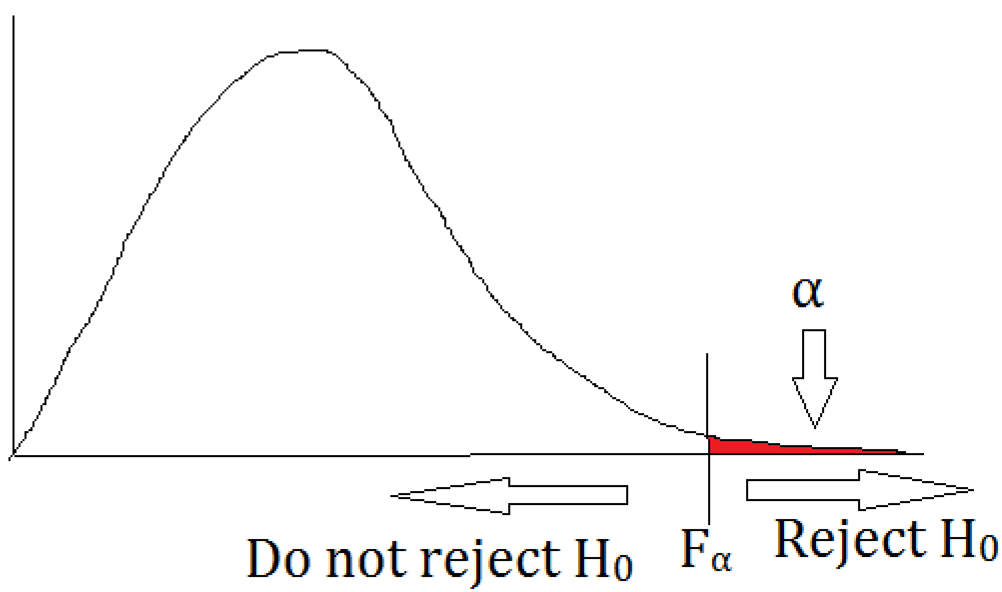
Interpreting F statistic

Decision Rule :

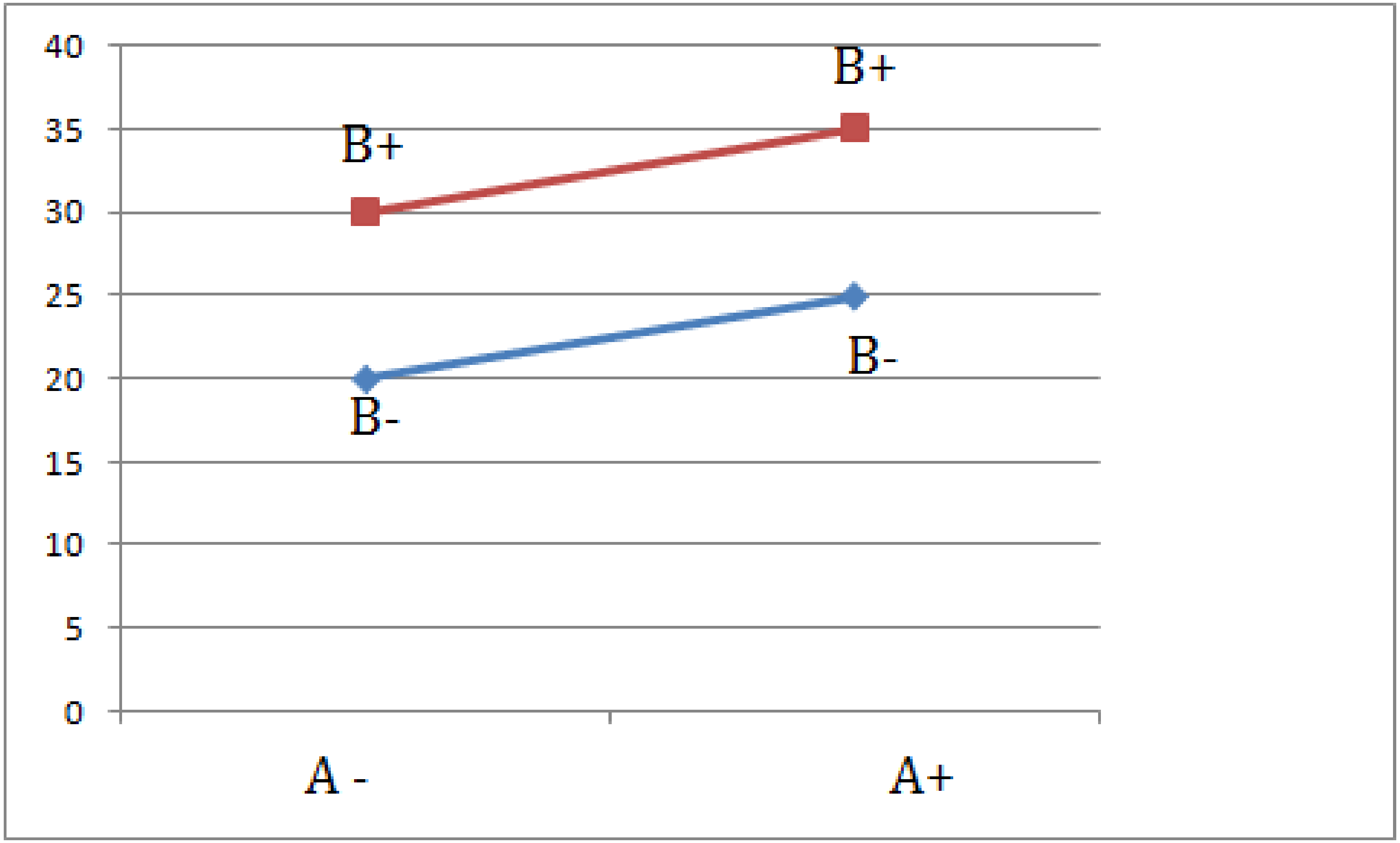
Reject H_0 if $F_0 > F_{\alpha}$

F_0 of A and F_0 of AB interaction are greater than F* and hence Factor A and Interaction of AB have significant effect or are influential in deciding the response

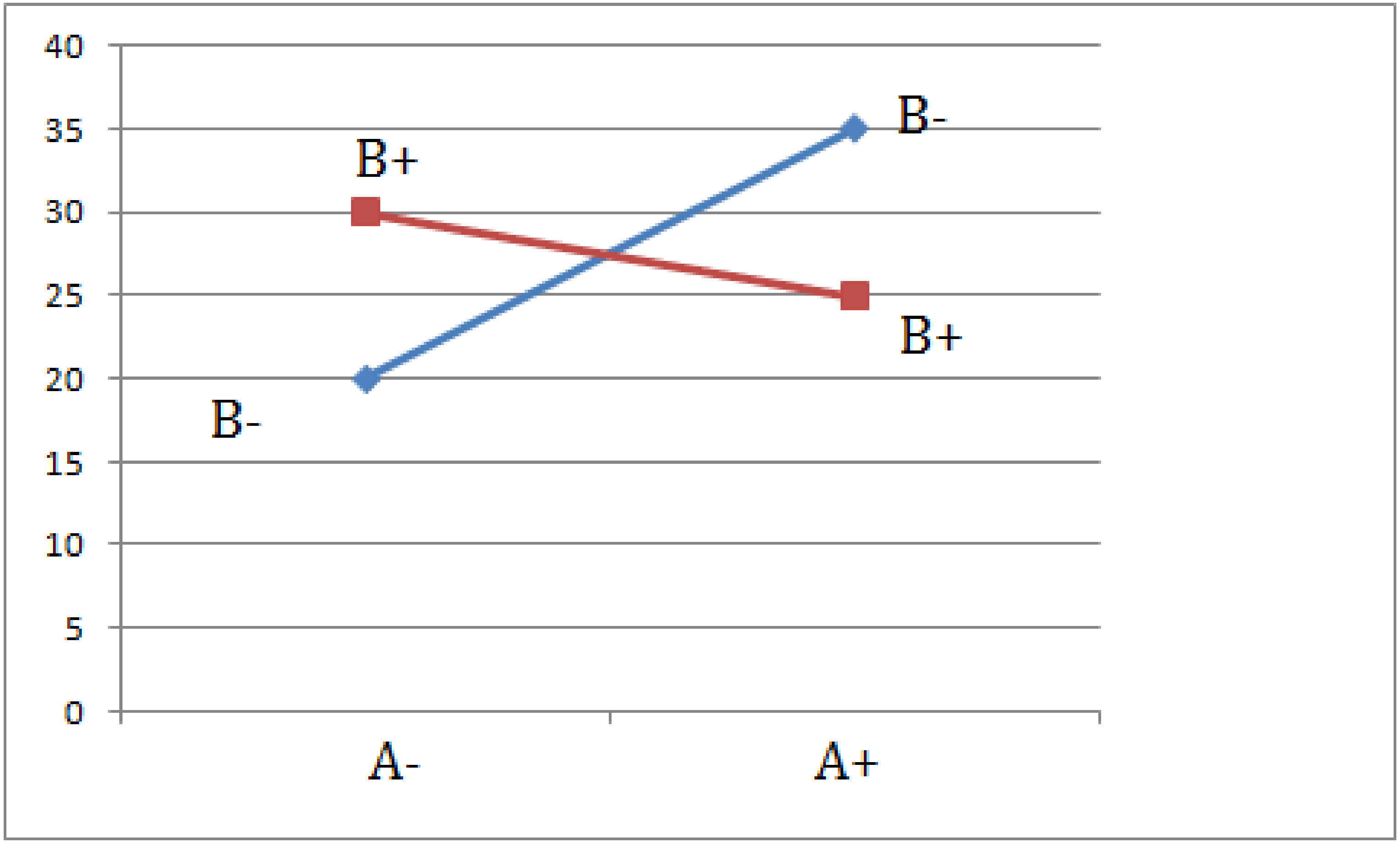
F Distribution



(1)	20
a	25
b	30
ab	35



(1)	20
a	35
b	30
ab	25



Example (15 marks)

			Replicates			
Trial No	A (hours)	B(^{0}C)	1	2	3	4
1	12	25	34	35	32	40
2	18	25	48	43	40	47
3	12	45	30	28	32	31
4	18	45	70	75	65	77

1. Estimate the factor effects. Which effects appear to be large?
2. Use analysis of variance to confirm your conclusions in part 1 taking p value of 0.05
3. Based on the analysis what levels of reaction time and temperature will you suggest for higher the better response.

Treatment	1	2	3	4	Total
(1)	34	35	32	40	141
a	48	43	40	47	178
b	30	28	32	31	121
ab	70	75	65	77	287
					727

Source of Variation	Effect
A	25.375
B	11.125
AB	16.125

Contrast = (Sum of responses at High) – (Sum of responses at Low)

Effect= (Contrast) / (r * 2^{k-1}) for 2^k factorial design (k : no. of factors)

Sum of Squares: $SS_{A/B/AB}$

$$= \frac{(\text{Contrast})^2}{r \times 2^k}$$

$$SS_T = \left\{ \sum \sum X_{ij}^{2} \right\} - (T^2/N)$$

$$SS_E = SS_T - (SS_A + SS_B + SS_{AB})$$

Mean Square: MS

$$= \frac{SS}{DOF}$$

$$F_0 = MS/MS_E$$

ANOVA

Source of Variation	SS	DOF	MS= SS/DOF	F_0= MS/MS_E
A	2575.6	1	2575.6	180.49
B	495	1	495	34.7
AB	1040	1	1040	72.89
Error	171.25	12	14.27	
Total	4281.9	15	285.46	

PART B

a) An industrial engineer employed by a beverage bottler is interested in studying the effects of two types of one litre deviations at the neck. The two bottles types (A) are glass and plastic. Two workers are used to perform a task of operating the same machine in different shifts. Four replicates of a 2^2 factorial design are performed and the deviations from the target observed are listed below in the following table. Estimate the factor and interaction effects. Perform an ANOVA and identify the significant factors. Which bottle type and which worker would you recommend? (15)

Bottle Type	Worker			
	1		2	
Glass	0.12	0.89	0.65	0.24
	0.98	0	0.49	0.55
Plastic	0.95	0.43	0.28	0.91
	0.27	0.25	0.75	0.71

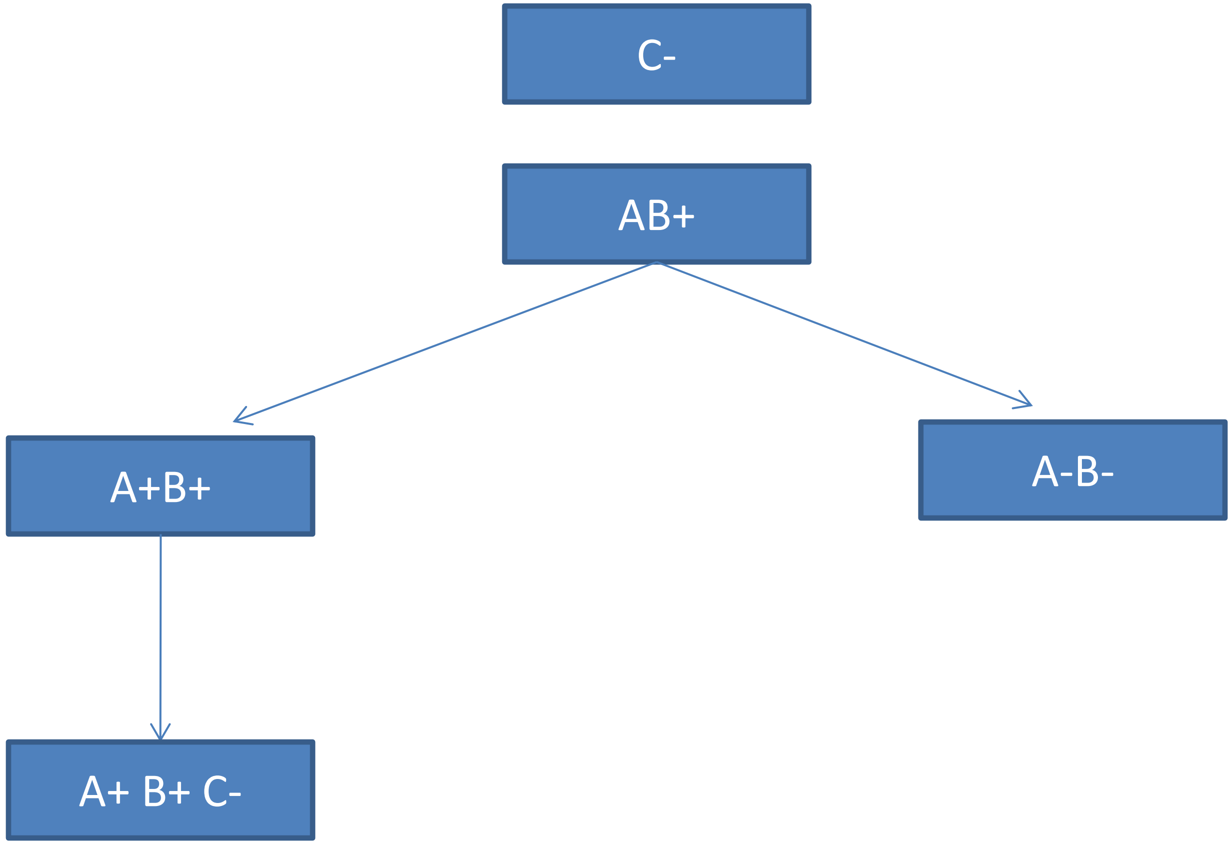
2^3 factorial design

	A	B	C	AB	AC	BC	ABC	Trtmnt.
1	-	-	-	+	+	+	-	(1)
2	+	-	-	-	-	+	+	a
3	-	+	-	-	+	-	+	b
4	+	+	-	+	-	-	-	ab
5	-	-	+	+	-	-	+	c
6	+	-	+	-	+	-	-	ac
7	-	+	+	-	-	+	-	bc
8	+	+	+	+	+	+	+	abc

Treatment	1	2	3	Total
(1)	6	8	9	23
a	10	16	15	41
b	18	12	15	45
ab	12	9	10	31
c	20	26	29	75
ac	34	28	32	94
bc	36	44	46	126
abc	25	22	24	71

Source of Variation	Effect
A	-2.667
B	3.333
C	18.833
AB	-8.833
AC	-3.333
BC	1.333
ABC	-3.500

Source of Variation	SS	DOF	MS= SS/DOF	F_0= MS/MS_E
A	42.66	1	42.66	4.023
B	66.66	1	66.66	6.299
C	2128.167	1	2128.167	201.093
AB	468.167	1	468.167	44.238
AC	66.667	1	66.667	6.299
BC	10.667	1	10.667	1.008
ABC	73.500	1	73.500	6.945
Error	169.33	16	10.583	
Total	3025.833	23		


C-
AB+
A+B+
A-B-
A+ B+ C-